

Laser Water Chiller: Compressor Not Running

BERT'S SHEET METAL PRODUCTS

Tongfei TFLW-3000WDR | 3-phase 380 V 60 Hz | 5.5 kW | 13.4 A | R134a 1.9 kg | S/N 1008240192145

Symptom: fan runs, water not cooling, compressor does not run. Water replaced, dust blown out.

Do not run the laser without cooling.

Date: _____ Checked by: _____

PART 1: SHOP CHECKS

no tools, panel closed

AT THE CONTROLLER

- ☐ Write down any **alarm code**.
- ☐ Note **water temp vs. setpoint**. The compressor only runs if the water is warmer than the setpoint.
- ☐ Which circuit is warm: **laser, optics, or both?**
- ☐ Power off, wait **5 min**, power on, wait 5 more (start delay). Compressor: **starts / hums / silent?**

REFRIGERANT GAUGES (UNIT OFF 30+ MIN)

- ☐ Read both gauges, write the psi below.
About 70-85 psi on both = refrigerant OK, likely electrical: go to Part 2.
Near 0 or low = refrigerant leak: call an **EPA 608 refrigeration tech**, not the electrician.
- ☐ Flashlight check for **oily spots** on copper joints and around the compressor (marks a leak).

WATER AND AIRFLOW

- ☐ Water level OK, **distilled / deionized** water.
- ☐ Water **filter** clean.
- ☐ **Condenser fins** clean; nothing blocking air in or out.
- ☐ Room not very hot (chillers struggle above ~95 F).

RECENT CHANGES

- ☐ Any **electrical work**, machine **moved**, or power feed **reconnected?** Swapped phases stop the compressor.
- ☐ Any **outages, storms or brownouts** before it stopped?

PART 2: ELECTRICIAN

live voltage, LOTO when open

SUPPLY

- ☐ Find the **step-up transformer** feeding 380 V / 60 Hz. Check its **input and output fuses / breaker**.
- ☐ Measure **all 3 phases at the chiller input** while running: ~380 V, balanced within a few percent.
- ☐ Look for **loose or burnt lugs** at the transformer and chiller.

CHILLER ELECTRICAL BOX

- ☐ **Phase-sequence / phase-loss protector**: LED status, tripped? Wrong phase order: **swap two input legs** at the supply.
- ☐ **Compressor breaker / fuses**: tripped or blown?
- ☐ **Thermal overload relay**: tripped flag? **Reset once** only. Note if and how fast it trips again.
- ☐ **Compressor contactor**: coil gets voltage and pulls in on a call for cooling? Contacts pitted or burnt?
- ☐ **High / low pressure switches**: open or closed? Some have a manual reset.

AT THE COMPRESSOR

- ☐ Contactor in: **voltage on all 3 legs** at the compressor terminals?
- ☐ Power off, leads off: **winding resistance** T1-T2, T2-T3, T1-T3 should be **about equal**.
- ☐ **Insulation to ground** from each terminal: should read **open (megohms)**. Short = bad compressor.
- ☐ If it starts: **running amps per leg**, balanced and near the 13.4 A nameplate rating.

Refrigerant work (leak repair, recharge, compressor swap) legally needs an EPA 608 certified tech.

READINGS

write them here and send a photo of this page

Alarm code	Water temp / setpoint	Compressor: start / hum / silent
Gauge 1 psi (off)	Gauge 2 psi (off)	Room temp F
Volts L1-L2	Volts L2-L3	Volts L1-L3
Ohms T1-T2	Ohms T2-T3	Ohms T1-T3
Insulation to ground	Amps L1 / L2 / L3	Tripped / reset (what)

Notes: _____